

OLS Error Assumptions in the Bigger Picture

扩展阅读：从 residual diagnostics 到 errors-in-variables 与 instrumental variables

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1 Why This Note Exists

这份扩展阅读的目标是把我们讨论过的 OLS 误差假设放进更大的图景里。核心问题是：

$$y = \Phi\theta + \epsilon$$

这里的关键不是“误差来自模型还是测量”，而是：

标准 OLS 把误差建模为加在 y 上，而不是加在 Φ 上。

也就是说，OLS 默认把 Φ 当作可信的解释结构，然后用 ϵ 吸收观测输出、模型不完备、随机扰动等造成的差异。

这份 note 主要回答三个问题：

1. OLS 在统计建模工具箱中的位置是什么？
2. residual diagnostic plots 检查的到底是哪一层假设？
3. errors-in-variables 与 instrumental variables 分别解决什么问题？

2 Big Picture Flowchart

下面的框图可以理解为一个诊断路线图。

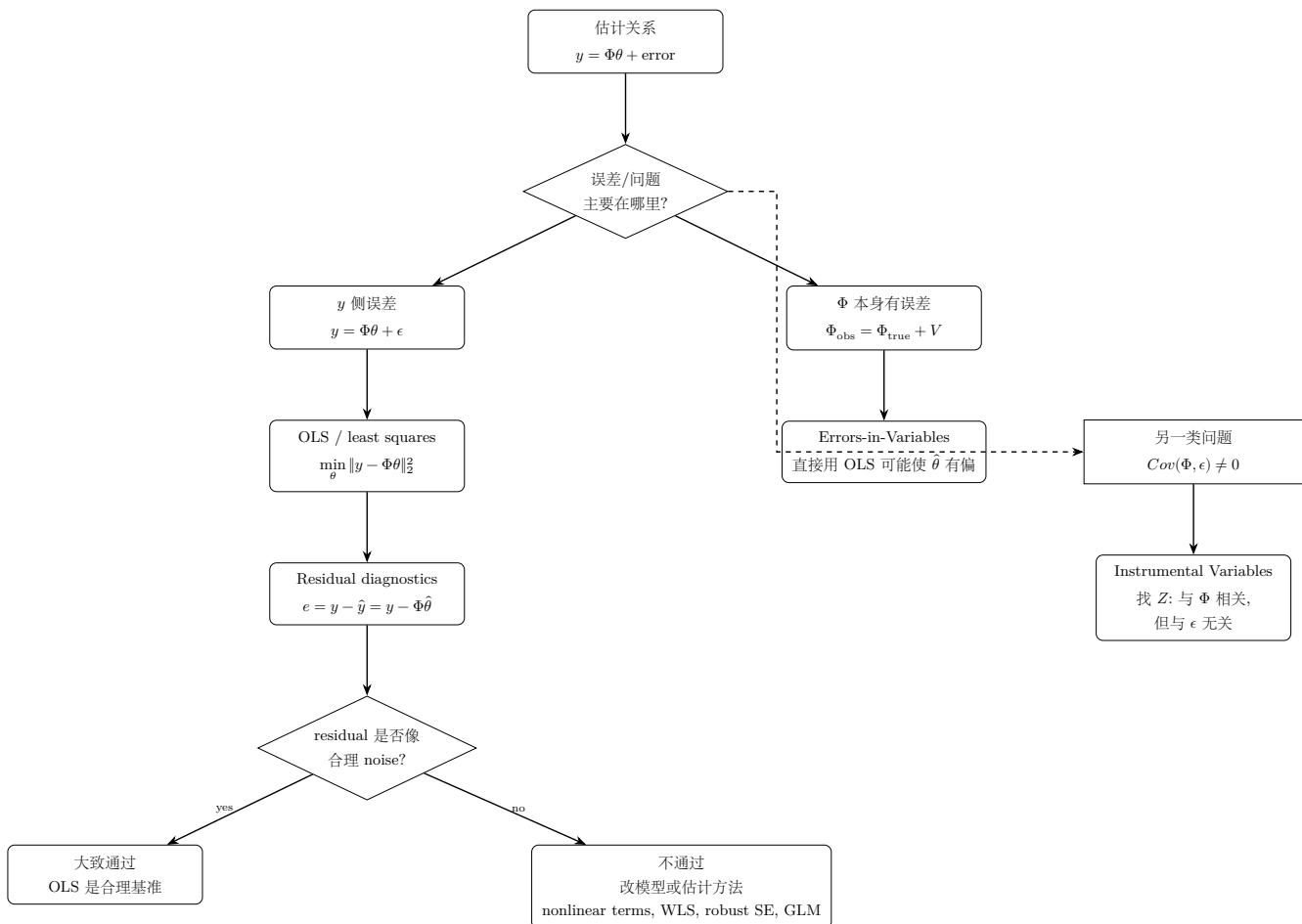


图 1: OLS、errors-in-variables 与 instrumental variables 的定位图

3 Mindmap: Where OLS Sits

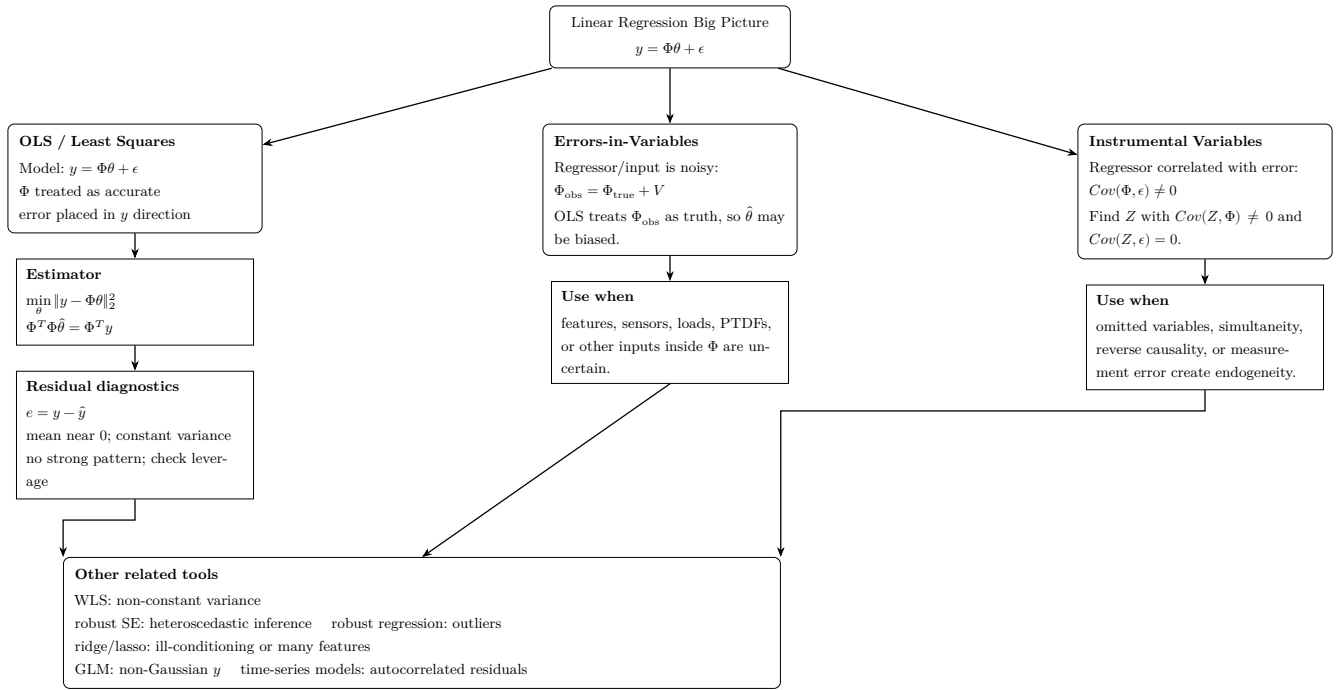


图 2: OLS 在更大线性回归工具箱中的位置

4 OLS: The Baseline Assumption

OLS starts from:

$$y = \Phi\theta + \epsilon$$

and estimates θ by solving:

$$\hat{\theta} = \arg \min_{\theta} \|y - \Phi\theta\|_2^2$$

This is equivalent to finding the parameter vector that makes the residual vector as short as possible:

$$e = y - \hat{y} = y - \Phi\hat{\theta}$$

The important modelling assumption is:

Φ is treated as known and accurate.

Then the unexplained difference is placed in:

$$\epsilon$$

This ϵ can include measurement noise in y , unmodelled dynamics, random disturbances, or modelling simplifications. OLS does not require us to say that the error is only from measurement or only from modelling. It only says that the error is represented on the y side.

5 What Residual Diagnostics Check

Residual diagnostics happen after fitting OLS. The sequence is:

```
fit OLS
-> compute residuals e_i = y_i - y_hat_i
-> inspect whether residuals behave like the assumed noise
-> decide whether the model structure or error assumption is plausible
```

This connects directly to the STAT448 Lab 3 idea:

$$e_i = y_i - \hat{y}_i$$

The residual is the observed value minus the predicted value. Residual-based quantities then separate total variation into explained and unexplained parts. In particular:

$$R^2 = 1 - \frac{RSS}{TSS}$$

where:

$$RSS = \sum_i (y_i - \hat{y}_i)^2$$

So RSS measures unexplained variation in y after the model has done its best under the OLS criterion.

5.1 Main Diagnostic Questions

The common residual plots answer different questions:

Diagnostic	What it asks	Warning sign
Residuals vs fitted	Is the linear form missing structure?	curve, trend, funnel shape

Diagnostic	What it asks	Warning sign
Scale-location	Is residual variance roughly constant?	variance grows or shrinks with fitted value
Q-Q residuals	Are residuals approximately normal?	strong tail deviation
Residuals vs leverage	Are a few points dominating the fit?	high leverage or high Cook's distance
Residuals vs each predictor	Is there predictor-specific nonlinearity?	systematic residual pattern against a predictor

The phrase “mean zero residuals” needs a small qualification. In OLS with an intercept, fitted residuals sum to zero by construction:

$$\sum_i e_i = 0$$

But this does not automatically prove the model is good. A bad model can still have residuals with average zero while showing curvature, heteroscedasticity, autocorrelation, or outlier influence.

6 When Residual Diagnostics Fail

If the residual plot shows a systematic curve, the issue may be model form:

```
linear model too simple
-> add nonlinear terms
-> add interactions
-> transform variables
-> use a different model class
```

If residual variance grows with fitted values, the issue may be non-constant variance:

```
heteroscedasticity
-> weighted least squares
-> robust standard errors
-> variance-stabilising transform
```

If there are influential points:

```
outliers or high leverage
-> inspect data quality
-> robust regression
-> sensitivity analysis
```

If residuals are autocorrelated:

```
time dependence
-> time-series model
-> dynamic model
-> correlated-error model
```

If Φ itself is noisy or endogenous, residual plots alone may not be enough. That is where errors-in-variables and instrumental variables become relevant.

7 Errors-in-Variables

The standard OLS model treats Φ as clean:

$$y = \Phi\theta + \epsilon$$

An errors-in-variables model says the observed regressor matrix is itself noisy:

$$\Phi_{\text{obs}} = \Phi_{\text{true}} + V$$

while the true relation is:

$$y = \Phi_{\text{true}}\theta + \epsilon$$

The problem is that we only observe Φ_{obs} , not Φ_{true} . If we directly run OLS using Φ_{obs} , then the estimator acts as if the noisy input were exact.

In a simple one-regressor case:

$$y = x_{\text{true}}\theta + \epsilon$$

but we observe:

$$x_{\text{obs}} = x_{\text{true}} + v$$

Direct OLS with x_{obs} often produces attenuation bias: the estimated slope is pulled toward zero. Intuitively, the horizontal coordinate is also shaking, but OLS only corrects vertical errors.

Errors-in-variables is appropriate when:

- input measurements are noisy;

- features are estimated from another model;
- sensor measurements affect Φ ;
- line parameters, PTDF values, loads, or other regressors are uncertain;
- the uncertainty in Φ is not negligible compared with the uncertainty in y .

8 Instrumental Variables

Instrumental variables address a related but slightly different issue: the regressor is correlated with the error term.

For a simple model:

$$y = x\theta + \epsilon$$

OLS needs:

$$\text{Cov}(x, \epsilon) = 0$$

If this fails, x is called endogenous. Common causes include:

- omitted variables;
- reverse causality;
- simultaneity;
- measurement error in x ;
- feedback between the system response and the explanatory variable.

IV looks for a variable z that satisfies two requirements:

$$\text{Cov}(z, x) \neq 0$$

and:

$$\text{Cov}(z, \epsilon) = 0$$

In words:

- z must be relevant: it explains variation in x ;
- z must be exogenous: it does not directly contaminate the error term.

A common implementation is two-stage least squares:

Stage 1:

use z to predict the problematic regressor x

Stage 2:

regress y on the predicted, cleaner part of x

In formula form:

$$x = z\pi + v$$

then:

$$y = \hat{x}\theta + \epsilon$$

The intuition is that IV does not trust all variation in x . It only uses the part of x that can be traced to a valid external source z .

9 DCOPF / Power-System Intuition

For a power-system estimation problem, the same distinction is useful.

Suppose:

$$y = \Phi\theta + \epsilon$$

where y might represent observed prices, flows, dispatch outcomes, or another dependent variable, while Φ contains explanatory structure such as loads, sensitivities, PTDF-like quantities, network parameters, or constraint indicators.

The OLS-style assumption is:

Φ is trusted enough to be treated as fixed.

The mismatch between observed y and predicted $\Phi\theta$ is placed into ϵ .

This is often a useful first model. It gives a transparent baseline and residuals that can be inspected.

But if the inputs inside Φ are themselves uncertain, the problem changes:

loads uncertain

line parameters uncertain

constraint status misclassified

PTDF or sensitivity matrix approximated

data generated by another noisy estimation step

Then the error is no longer purely a y -side error. The model is closer to:

$$y = (\Phi + \Delta\Phi)\theta + \epsilon$$

Expanding:

$$y = \Phi\theta + \Delta\Phi\theta + \epsilon$$

Now the residual contains:

$$\Delta\Phi\theta$$

This term depends on the parameter being estimated. That is why the problem is harder: the error is entangled with the regressor and the unknown coefficient.

10 Choosing the Tool

Situation	First tool to think about	Why
Φ is trusted, noise is mainly in y	OLS	baseline estimator, simple residual interpretation
residuals show curvature	add nonlinear terms or transform variables	model form is too simple
residual variance changes with fitted value	WLS or robust standard errors	OLS coefficients may still be useful, but inference changes
residuals have strong outliers	robust regression or sensitivity analysis	squared loss over-weights large residuals
Φ is measured with error	errors-in-variables	OLS treats noisy regressors as truth
Φ is correlated with ϵ	instrumental variables	OLS no longer isolates causal/structural variation
many correlated predictors	ridge / lasso	controls variance or performs selection
response is binary, count, or bounded	GLM	ordinary Gaussian-error LS may be wrong
residuals are time-correlated	time-series / dynamic model	independence assumption fails

11 Key Takeaways

1. OLS is the baseline method for:

$$y = \Phi\theta + \epsilon$$

where Φ is treated as fixed and reliable.

2. Residual diagnostics do not mainly check whether OLS solved the optimisation problem correctly. They check whether the fitted residuals look compatible with the assumptions behind the model.
3. If residuals show systematic patterns, the model may be missing nonlinear terms, interactions, variables, or a different error structure.
4. Errors-in-variables is needed when Φ itself is noisy:

$$\Phi_{\text{obs}} = \Phi_{\text{true}} + V$$

5. Instrumental variables is needed when the regressor is not independent of the error:

$$\text{Cov}(\Phi, \epsilon) \neq 0$$

6. In engineering terms, OLS says:

I trust the explanatory structure enough to fit vertical residuals.

Errors-in-variables says:

I do not fully trust the explanatory structure itself.

Instrumental variables says:

The explanatory variable is contaminated by something related to the residual, so I need an external source of clean variation.